

WASP-10b

R-0894

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_240705 · created 2026-07-29T02:43:01+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460496.9127 +/- 0.0064 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.06
Transit depth (Rp/Rs)^2	3.81 +/- 2.33 [%]
Orbital Inclination (inc)	85.84 +/- 1.89
Airmass coefficient 1 (a1)	3.06 +/- 1.99
Airmass coefficient 2 (a2)	-0.91 +/- 0.81
Scatter in the residuals of the lightcurve fit is	68.72 %
Best Comparison Star	#3 - [479, 338]
Optimal Aperture	3.81
Optimal Annulus	5.57
Transit Duration (day)	0.063 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.06 vs 0.1592 (deviation 0.59σ)
Duration fit vs published	0.063 d vs 0.0946 d (deviation 0.97σ)
Mid-time O–C	-14.8 min
Depth SNR	3.2
Residual scatter	68.72 %

Light curve

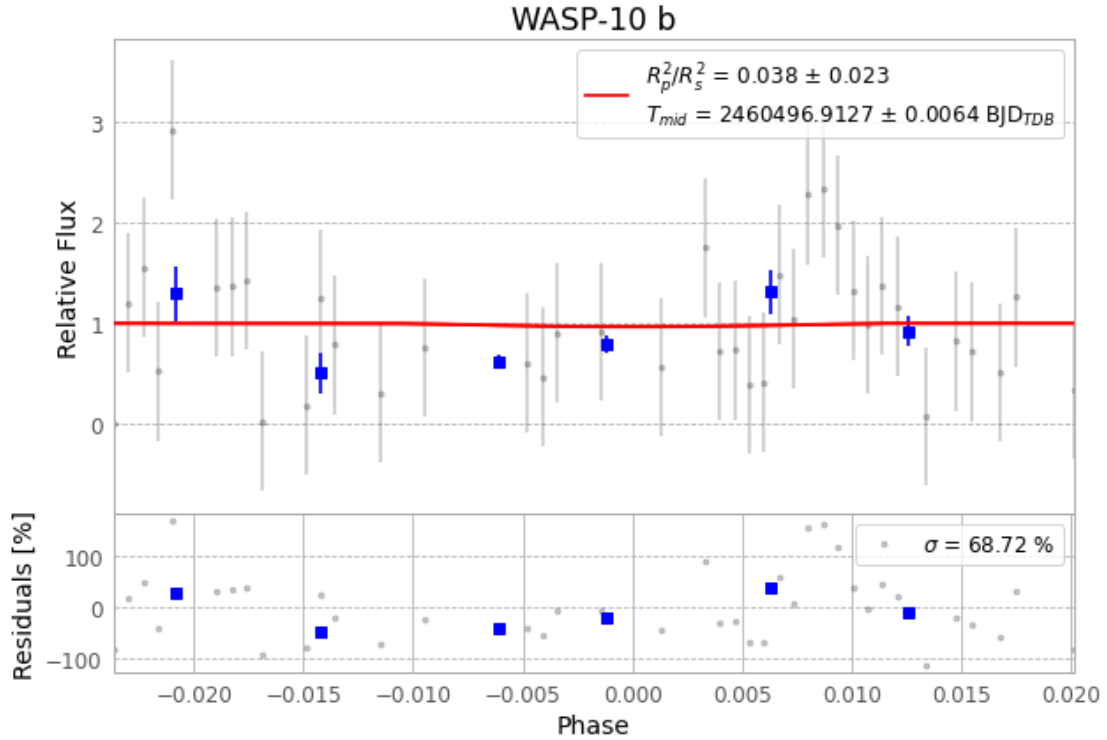


Figure 1 — FinalLightCurve_WASP-10 b_2024-07-05.png (SHA-256 d865a844ea879b4f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [290, 48], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 394a5a58541bc5e7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

66 FITS frames (43 MB), WASP-10240705080615.FITS → WASP-10240705112115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-240705.log (8ae9f5bde1e7...), FinalLightCurve_WASP-10 b_2024-07-05.png (d865a844ea87...), FinalLightCurve_WASP-10 b_2024-07-05.csv (6ba05684406c...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 02:43Z.